

# Comparative Study: 8086 to Pentium 4 — MPCA Module 6

**Definition:** The evolution of the **Intel x86 Microprocessor Architecture** from the 16-bit 8086 to the 7th-generation 32/64-bit Pentium 4 demonstrates radical advancements in bus widths, pipeline depths, caching hierarchies, and parallel instruction execution.

## 1. Comprehensive Master Comparison Table

| Feature / Processor | Intel 8086         | Intel 80386DX      | Intel Pentium (P5)         | Intel Pentium II                | Intel Pentium 4                         |
|---------------------|--------------------|--------------------|----------------------------|---------------------------------|-----------------------------------------|
| Year Introduced     | 1978               | 1985               | 1993                       | 1997                            | 2000                                    |
| Data Bus Width      | 16-bit             | 32-bit             | 64-bit                     | 64-bit                          | 64-bit                                  |
| Address Bus Width   | 20-bit             | 32-bit             | 32-bit                     | 36-bit (PAE)                    | 36-bit (64 GB)                          |
| Physical Memory     | 1 MB               | 4 GB               | 4 GB                       | 64 GB                           | 64 GB                                   |
| Virtual Memory      | None               | 64 TB              | 64 TB                      | 64 TB                           | 64 TB                                   |
| Clock Frequency     | 5 - 10 MHz         | 16 - 33 MHz        | 60 - 200 MHz               | 233 - 450 MHz                   | 1.3 - 3.8 GHz                           |
| Pipeline Depth      | 2-stage (BIU/EU)   | 3-stage            | 5-stage (Superscalar)      | 14-stage (Dynamic)              | 20 to 31 stages (Hyper-Pipelined)       |
| Integer Execution   | 1 inst / ~4 clocks | 1 inst / 2 clocks  | 2 inst / clock (U/V)       | 3 micro-ops / clock             | Out-of-Order Engine                     |
| L1 Cache            | None               | None               | 16 KB (8K I + 8K D)        | 32 KB (16K I + 16K D)           | 8 KB D-Cache + 12K micro-op Trace Cache |
| L2 Cache            | None               | External           | External                   | 512 KB (Cartridge)              | 256 KB - 2 MB On-die                    |
| Key Innovations     | Segmented memory   | 32-bit MMU, Paging | Superscalar, Dual U/V, BTB | Dual Independent Bus (DIB), MMX | NetBurst, Hyper-Threading, Trace Cache  |

## 2. Key Architectural Milestones

- 1. **8086:** Introduced 16-bit computing and segmented memory architecture.

2. **80386:** Introduced 32-bit processing, flat memory model, Protected mode with 4-level privilege rings, and 2-level Paging MMU.
  3. **Pentium:** Introduced superscalar dual integer pipelines (U and V pipes) and dynamic branch prediction.
  4. **Pentium 4:** NetBurst hyper-pipelined microarchitecture, Execution Trace Cache, and Hyper-Threading Technology.
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### 3. Quick Recall Flow

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8086 (16-bit, 1MB) -> 80386 (32-bit, 4GB, Paging) -> Pentium (Superscalar U/V, 64-bit bus) -> Pentium 4 (NetBurst, Hyper-Threading)